

Chapter 6

Knowledge Profiles

A curriculum can look complete on paper. Courses are listed. Credits are distributed. Outcomes are written. Assessments are mapped. Yet an important question often remains unasked: *What kind of knowledge are we actually building in our students?*

- ⇒ *Are we strengthening their foundation in mathematics and science?*
- ⇒ *Are we developing their ability to apply engineering principles in practical situations?*
- ⇒ *Are we exposing them to complexity, uncertainty, and multidisciplinary thinking?*

Knowledge Profiles help answer these questions.

In engineering education, knowledge is not one-dimensional. It ranges from fundamental theory to applied practice, from routine procedures to complex problem-solving in uncertain environments. A balanced program must intentionally develop different categories of knowledge rather than focusing narrowly on one aspect.

Knowledge Profiles provide a structured way to classify and evaluate the depth and nature of learning within a course or program. They ensure that outcomes are not only measurable, but academically rigorous and professionally relevant. By

mapping courses against defined knowledge categories, faculty can identify gaps, avoid overemphasis on routine content, and maintain alignment with engineering standards.

This chapter introduces the concept of Knowledge Profiles and explains how they guide curriculum design, outcome mapping, and quality assurance within an Outcome-Based Education framework.

6.1 Introduction to Knowledge Profiles

Knowledge Profiles describe the **types** and **depth** of **knowledge** that engineering students are expected to acquire during a program. In Outcome-Based Education, Knowledge Profiles help ensure that Program Outcomes are supported by appropriate theoretical, practical, and professional knowledge.

BAETE uses Knowledge Profiles to clearly specify the depth, breadth, and nature of knowledge required to address complex engineering problems. These profiles help identify whether a problem demands fundamental engineering knowledge, specialist expertise, practical experience, or research-based understanding, ensuring that the level of complexity is appropriate and aligned with program outcomes.

6.2 Purpose of Knowledge Profiles in OBE

Knowledge Profiles are used to:

- Classify different types of engineering knowledge
- Support Program Outcomes
- Ensure balanced curriculum design
- Guide assessment and accreditation

They help faculty understand what kind of knowledge is being developed through courses and activities.

6.3 Knowledge Profile Categories (K1–K8)

BAETE defines eight Knowledge Profile categories, labeled K1 to K8. Each category represents a specific type of knowledge.

Table 6.1: Description of Knowledge profiles

Attribute	Title	Description
K1	Natural Science Knowledge	A systematic, theory-based understanding of the natural sciences applicable to the discipline.
K2	Mathematics and Computational Foundations	Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline.
K3	Engineering Fundamentals	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
K4	Engineering Specialist Knowledge	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
K5	Engineering Design Knowledge	Knowledge that supports engineering design in a practice area.
K6	Engineering Practice and Technology	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
K7	Comprehend (Engineering, Society, Ethics, and Sustainability)	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability.

K8	Research and Scholarly Knowledge	Engagement with selected knowledge in the research literature of the discipline.
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6.4 Importance of Knowledge Profiles

Knowledge Profiles are important because they:

- Ensure coverage of both theoretical and practical knowledge
- Support mapping of Program Outcomes to curriculum
- Help justify PO attainment during accreditation
- Promote balanced development of graduates

They also help identify gaps where certain types of knowledge may be underemphasized.

6.5 Knowledge Profiles and Program Outcomes

According to BAETE, each Program Outcome (PO-1 to PO-8) is linked to one or more Knowledge Profiles. For example:

Table 6.2: Mapping of POs with Knowledge Profiles

PO No.	Program Outcome Title	K1	K2	K3	K4	K5	K6	K7	K8
PO 1	Engineering Knowledge	✓	✓	✓	✓				
PO 2	Problem Analysis	✓	✓	✓	✓				
PO 3	Design / Development of Solutions					✓			
PO 4	Investigation								✓
PO 5	Modern Tool Usage						✓		
PO 6	The Engineer and Society							✓	
PO 7	Environment and Sustainability							✓	
PO 8	Ethics							✓	

This linkage ensures coherence between outcomes and curriculum content.

6.6 Mapping Knowledge Profiles with Course Outcomes

Mapping Knowledge Profiles with Course Outcomes ensures that a course develops not only the correct cognitive depth but also the appropriate type of engineering knowledge. While Bloom's Taxonomy defines *the level of thinking*, Knowledge Profiles define the nature and complexity of knowledge expected in an engineering program.

How to Map Knowledge Profiles with COs

- **Begin with the Course Intent:** Clarify the primary purpose of the course. Is it focused on foundational theory, applied techniques, system design, experimentation, or solving complex engineering problems?
- **Examine Each Course Outcome Carefully:** For every CO, determine the dominant knowledge dimension it addresses.
 - Is it grounded in basic mathematics or scientific principles?
 - Does it apply engineering methods to routine problems?
 - Does it involve analysis of complex or unfamiliar situations?
 - Does it integrate social, environmental, or professional considerations?
- **Align COs with Relevant Knowledge Profile Categories:** Assign the appropriate Knowledge Profile category (such as K1–K8, depending on the adopted framework). The mapping should reflect the core knowledge requirement of the outcome.

- **Ensure Logical Consistency:** The mapped category must match the actual teaching–learning activities and assessment methods. Mapping should be defensible and supported by evidence.

Important Consideration: Alignment with Program Outcomes

In many accreditation frameworks, including those guided by the Board of Accreditation for Engineering and Technical Education (BAETE), certain Knowledge Profile expectations are predefined at the Program Outcome (PO) level. Since Course Outcomes are mapped to POs, the Knowledge Profile associated with a PO must also be reflected in the corresponding COs.

In other words:

If a particular PO is defined under a specific Knowledge Profile category, any CO mapped to that PO should logically align with the same knowledge expectation. This ensures vertical consistency from course level to program level and maintains compliance with accreditation standards.

Thumb Rules for CO–Knowledge Profile Mapping

- **One CO, One Dominant Knowledge Category:** Each CO should primarily reflect one main Knowledge Profile area. Secondary mapping may be possible but should remain justified.
- **Respect Course Level Progression:** Lower-level courses typically emphasize foundational knowledge, while senior and capstone courses should address applied, integrated, and complex knowledge dimensions.

- **Maintain Bloom's and Knowledge Alignment:** The cognitive level and knowledge category must support each other. Higher-order cognitive outcomes should not be mapped only to elementary knowledge categories.
- **Ensure Program-Level Coherence:** CO mappings should collectively support the overall Knowledge Profile distribution required by the program and accreditation body.

Effective Knowledge Profile mapping strengthens curriculum coherence, ensures compliance with accreditation expectations, and guarantees that students develop the right depth and type of engineering knowledge throughout their academic journey.

6.7 Summary

Knowledge Profiles define the knowledge foundation required for achieving Program Outcomes in engineering education. By integrating K1 to K8 across courses and assessments, programs can ensure that graduates are well-prepared to solve complex engineering problems and meet BAETE accreditation requirements.